

detect finger and hand movement is through fiber optics. Light is passed through the connected cable and is detected by a sensor at the other end. If the hand is curled up as a fist, less light reaches the sensor. This approach was implemented originally by DataGlove.



The NES PowerGlove also utilized optical sensors to detect finger flexing

Another approach also used is using strips of flexible conductive material. As the user curls his hand, resistance across a circuit changes. This change is detected by the CPU and the VR environment responds accordingly. This approach is less accurate compared to fiber optics.

On the other hand, gloves and shoes can also be used as output devices to provide the user haptic feedback, or feedback that they can feel. If you think this would require some extremely high level of technology, think of the vibrating devices inside your smartphones that have been doing this for years. And as all smartphone components, this has been getting smaller, cheaper and less power consuming each year.

To provide haptic feedback, you need a microcontroller, a vibrating actuator, i.e. a motor that can move in response to external commands, and a small power source like a battery. Various microcontrollers have been detailed extensively in our last Fastrack issue, and we would leave the choice to you to choose the correct one. Such microcontrollers should be programmable, be able to interface with sensors and actuators effectively and should have wireless capabilities that allow it to respond to commands sent from a smartphone, or the external system controlling the VR experience. The Gloveone VR glove contains as much as 10 actuators per finger to allow the user to feel objects in the VR environment.